

Design and Mechanical Properties of Anchorage system with Variable Stiffness and Bending Angle for CFRP Cables: Experiment and Simulation

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Abstract

Owing to the shear and compressive properties of CFRP perpendicular to the fiber direction are much lower than the tensile properties along the fiber direction, which leads to the anchoring of CFRP is difficult and limits its application in engineering structures. In the present paper, a new type anchorage system with variable stiffness of bonding medium and bending angle of CFRP bar is developed. The design, mechanical properties, stress distribution and parameter optimization of anchorage are conducted through experiment and simulation. The results show that the optimal bending angle of CFRP bar is 4° for anchorage design. The stress distribution inside the anchorage zone obtained by the simulation analysis is consistent with experimental test, and the setting of variable stiffness of bonding medium and bending angle of CFRP bar can reduce the stress peak by 50.9%-67.3% in the loading end, which avoid the premature local shear and slip failure of anchorage, and the anchoring efficiency is greater than 95%. Uniformity of stress distribution improved with the increase of the number and the evenly of length of bonding media, and the optimal average modulus (E_a) and difference of modulus (ΔE) of bonding medium are 12 GPa and 4 GPa, which provide the key design parameter for anchorage.

Keywords: CFRP cables, variable stiffness, bending angle, experiment and simulation, anchoring efficiency.

1 Introduction

Cable structures have been widely used in long-span space structures, bridge structures, and marine mooring systems [1]. Currently, the cables are mostly made up of high-strength steel wire and steel strands. Due to the disadvantage of higher weight, terrible corrosion and fatigue resistance of steel cables, which limits the breakthrough of long-span and brings about the shorter service life of cable structures, the maintenance and replacement of steel cables will cause huge economic losses [2-4]. Carbon fiber reinforced polymer (CFRP) has the advantages of light weight, high strength, superior corrosion, creep and fatigue resistance, it is suitable used in bridge structures instead of steel cables [5, 6].

CFRP cables have been demonstrated and applied in number of engineering structures. The Tsukuba Bridge in Japan in 1996 is the first CFRP cable bridge in the world, and CFRP cables are composed of two types of CFRP bars [2, 7]. The Stork Bridge in Switzerland in 1996 is first highway cable-stayed bridge with CFRP cables in the world, which includes two CFRP cables, and each CFRP cable consists of 241 paralleled CFRP bars with a diameter of 5 mm [2, 8, 9]. The Neigles Footbridge in Switzerland in 1998 is the first suspension bridge with CFRP cable in the word [2, 10]. In addition, the United States, Japan, France and other countries have conducted demonstration applications of CFRP bridge cables. The pedestrian cable-stayed bridge of Jiangsu University in 2005 is the first CFRP bridge cables in China, and three different cable sizes of CFRP cable are used according to different bearing requirements [2, 11]. The Aizhai Bridge in Hunan province in 2010 adopted the CFRP bar with a diameter of 12.6 mm used for sling cables [2, 12]. In addition, more than ten bridges with CFRP cable have been applied in China. However, as a typical orthorhombic material, the shear and compressive properties of CFRP perpendicular to the fiber direction are much lower than the tensile properties along the fiber direction, which makes the anchor of CFRP cable difficult and limits its application in engineering structures. Therefore, the development of high efficiency anchorage for CFRP cables is the key problem to be solved urgently [13-15].

The existing CFRP cable anchorages mainly include three types: mechanical-type, bonded-type and composite-type [16]. Mechanical-type anchorage generally consists of CFRP cable, steel barrel and wedges. This anchorage depends on friction of anchorage to balance the tensile load [17]. In 1991, Kakiyama et al. [18] studied the anchoring performance of two-piece clip anchorages for CFRP bars experimentally, and the test results showed that its anchoring efficiency reached 97%. In 1998, Syed -Ahmed and Shrive et al. [19] proposed a new type anchorage system, which not only set a small angle difference between the steel barrel and wedge, but also set a soft sleeve between the steel wedge and FRP bar. In 2006, Al-Mayah et al. [20] proposed the concept of

variable curvature for steel barrel, which can provide a uniform contact pressure for CFRP bars. In 2003, Mahmoud M et al. [21] used ultra-high strength concrete material to replace steel barrel and wedge, and CFRP cloth was wrapped outside of the barrel to improve the crack resistance. The results showed that this anchorage has high load bearing capacity and superior corrosion resistance. In summary, mechanical-type anchorage has the advantage of high anchoring efficiency with small volume. However, the disadvantages of being prone to local shear failure and poor fatigue resistance limits its large-scale application in engineering structures.

Bonded-type anchorage generally consists of CFRP cable, steel barrel and bonding medium. CFRP cable is fixed in the steel barrel by the bonding medium, and the anchorage force consists of chemical bonding force, friction force and mechanical bite force [22, 23]. According to the difference of inner core angle of the steel barrel, the anchorage can be divided into straight barrel type and inner cone barrel type [24]. In 1989, Wolff and Miesser et al. [25] successfully anchored FRP bars using epoxy sand as a bonding medium. In 1992, Mochida et al. [26] adopted cement slurry as bonding medium, and the test results showed that the effect of anchoring length on bearing capacity of anchorage was significance. In 1996, Meier et al. [27] proposed an anchorage system that the elastic modulus of the bonding medium gradually decreases from the free end of the anchorage to the loading end, which reduces the stress concentration of CFRP cables, and the anchorage has been applied to Stork Bridge successfully. In summary, bonded-type anchorage has the advantage of avoiding local shear damage, which has been used in engineering structures widely. However, this anchorage has some shortcomings, such as large slip and insufficient bearing capacity owing to the addition of bonding medium.

Composite-type anchorage combines the features of mechanical and bonded-type anchorage. Jiang et al. [2, 28] proposed an anchorage system in which CFRP cable is bonded with a bonding medium and wrapped by wedges, and then the whole part was put into the anchor barrel. In addition, a composite-type anchorage system is used on Aizhai Bridges of China, CFRP cable is bonded by a bonding medium near the loading end and a wedge clamp near the free end of anchorage [29]. In conclusion, the above two types of composite-type anchorage have the advantages of high anchorage efficiency, however, the anchorage has the disadvantages of large volume and difficult to prepare [30].

In addition to experimental tests, a number of scholars have carried out finite element simulation analysis of anchorages [31, 32]. In 1974, Mitchel et al. [33] used an asymmetric model to analyze the bonded-type anchorage of FRP bars. The results showed that the shear stress peak of FRP bars can be reduced by the setting of variable stiffness bonding medium. In 1993, Holte et al. [34] analyzed the effect of different design parameters on bonded-type anchorage through finite element method. The results showed that the anchoring

performance of the anchorage improved with the increase of inner core angle of steel barrel. In 2001, Mayah A et al. [35] investigated the effect of the metal soft sleeve between FRP bars and steel wedges on the anchoring performance, the setting of metal soft sleeve improved the bearing capacity of anchorage. In conclusion, finite element simulation is a powerful method to obtain the stress distribution and transfer inside the anchorage, which can be used to optimize the parameters of the anchorage.

The current research about anchorage focuses on the design of bearing capacity, which lacks of the systematic study from material - structure design - parameter optimization. In the present paper, the mechanical properties of raw material for bonding medium and CFRP bar are obtained experimentally. Furthermore, the new type anchorage system with variable stiffness of bonding medium and bending angle of CFRP bar is developed. And then the design, mechanical properties, stress distribution and parameter optimization of anchorages are conducted through experiment and simulation.

2 Materials and Methods

2.1 Raw materials

The anchorage consists of four parts: CFRP bars, bonding medium, steel barrel and locating plate. The diameter and the fiber volume content of CFRP bars are 4 mm and 70%, respectively. The bonding medium is made of T_s epoxy resin mixed with SiC powder (800 mesh) with the mass ratios of 1:1, 1:1.2, 1:1.4, 1:6, 1:1.8 and 1:2. The locating plate and steel barrel are made of stainless steel.

2.2 Innovative design of anchorage

The bonding medium used in the conventional anchorages for CFRP cables is constant stiffness, which brings about the stress concentration of CFRP cable in the loading end of anchorage [36], resulting the premature failure of CFRP cable [37, 38]. The variable stiffness bonding medium is used for anchorage system in the present paper, six bonding media is distributed along the longitudinal direction of anchorage, and the elastic modulus of bonding media increases from the loading end to free end of anchorage.

In addition, the parallel CFRP bars is used in the conventional anchorages for CFRP cables, which brings about the significant slip between CFRP cable and steel barrel, reducing the reliability and security of CFRP cables [39, 40]. CFRP bars have a bending angle along the longitudinal direction of anchorage, which increases the contact area between CFRP bar and bonding medium, improving the mechanical biting force and reducing the slip between CFRP cable and anchorage significantly.

To ensure that the stress distribution of CFRP bars inside the anchorage is uniform, the central CFRP bar is removed from the anchoring region and the remaining CFRP bars are distributed in a diverging manner at the

same angle. Two kinds of CFRP cable ($\Phi 4-6$ and $\Phi 4-18$) are designed in the present paper, which consisted of six and eighteen CFRP bars, respectively. The length of free end of cable is 700 mm. The detailed anchorage parameters are listed in **Table 1**, and the schematic of anchorage system with six CFRP bars is shown in **Figure 1**.

In addition, in order to quantitatively analyze the performance improvement of innovative anchorage with variable stiffness and bending angle (VB) compared with conventional anchorage, three other anchors were also designed, which anchorage with equal stiffness and 0 angle (EP), equal stiffness and bending angle (EB), variable stiffness and 0 angle (VP). It should be noted that the mass ratio of bonding media of EP and EB was 1:2.

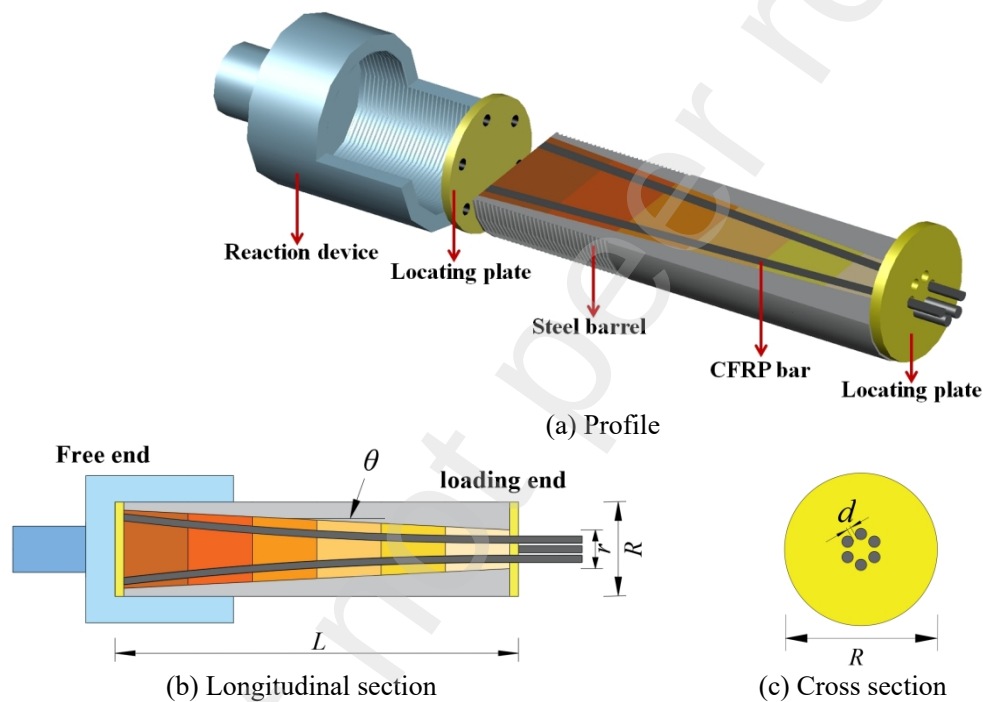


Figure 1. Schematic of the anchorage ($\Phi 4-6$)

Table 1. Parameters of the anchorage

Cable type	L (mm)	R (mm)	r (mm)	θ ($^{\circ}$)	d (mm)
$\Phi 4-6$	200	36	16	4	2
$\Phi 4-18$	300	51	28	4	2

2.3 Preparation of anchorage

The detailed assemble process of anchorage system is as follows:

1) Sandpaper was used to polish CFRP bars of anchorage zone. The locating plate was used to assemble number of CFRP bars into cable, then the cable was inserted into the steel barrel and CFRP bars were bent with 0° and 4° by the locating plate in anchorage zone. Subsequently, the strain gauges were pasted on four positions: both end and the middle of anchorage zone, middle of cable, and two strain gauges pasted on each position with the spacing of 5 mm.

2) The bonding medium is injected into the steel barrel using a syringe in the order of the elastic modulus from small to large. After the initial solidification of the bonding medium is completed at intervals of 6 h, the next pouring of the bonding medium is performed.

3) The heating belt was winded around the steel barrel, and the samples were cured at the temperature of 60°C for 48 h

The preparation process of the anchorage system is shown in **Figure 2**, and assembled anchorage system is shown in **Figure 3**.

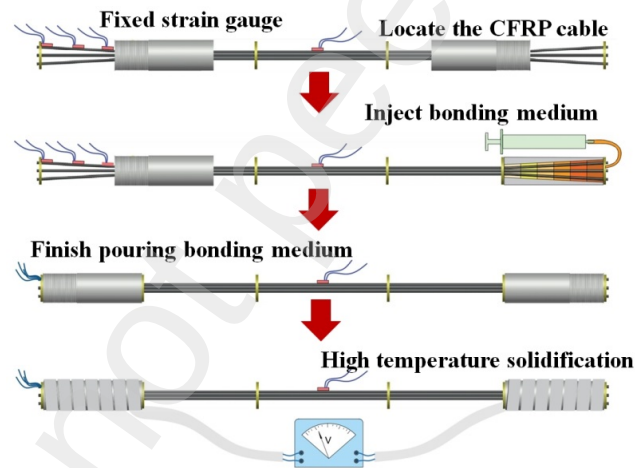


Figure 2. Preparation of the anchorage system.

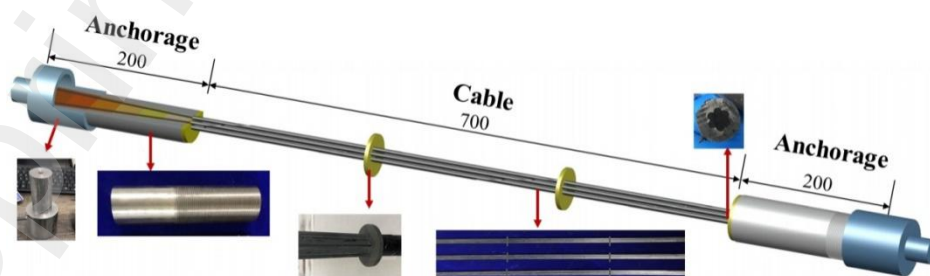


Figure 3. Assembled anchorage system (unit: mm).

2.4 Mechanical tests of raw materials

The mechanical properties of raw materials are the key parameters for the design of anchorage, which includes tensile strength, tensile elastic modulus and Poisson's ratio of CFRP bars with the bending angle of 0 °, 2 °, 4 °, 6 °, 8 ° and 10 °, compressive elastic modulus of CFRP bars, compressive strength, elastic modulus and Poisson's ratio of bonding medium. The universal testing machine (DHY-10080, Shanghai Hengyi Precision Instrument Co., Ltd, Shanghai) is used for the above mechanical property tests,

Tensile test of CFRP bars with bending angles of 0 °, 2 °, 4 °, 6 °, 8 ° and 10 ° were performed according to ASTM D7205/D7205M-06. The anchorage developed by the previous research was used for tensile test of CFRP bar, the anchoring length was 150 mm, and the free section length was 300 mm [41]. The displacement control mode was adopted with a loading rate of 2 mm/min to conduct the tensile test.

CFRP bar was cut into a cuboid with the length of 25 mm and cross section area of 6.25 mm² for compressive test. The displacement control mode with a loading rate of 1 mm/min was used to conduct the compressive test.

The bonding medium was made into a cube with a length of 40 mm for compressive testing. The strain along and perpendicular to the compression direction was measured by strain gauges. The displacement control mode with a loading rate of 2 mm/min was used to conduct the compressive test, and the test was stopped until the sample was crushed.

2.5 Tensile tests of anchorage

The tensile test of CFRP cable was conducted using a 100 t universal tensile machine (WAW-1000, Shanghai, China). Displacement control was selected with a loading rate of 5 mm/min. The strain gauge was arranged in the center of CFRP cable to record the change of strain. Additionally, a pretension equivalent to 15% of the ultimate load was applied to the sample at a controlled loading rate of 5 mm/min, followed by unloading to 0. The tension test was stopped when CFRP cable presented the burst, slip and local shear failure.

2.6 Finite element analysis

To deeply explore the bonding performance between the cable and the bonding medium, the finite element software "Abaqus" was used for simulation analysis [42]. To simplify the calculations, only one-sixth of the solid model was built. The steel barrel and the bonding medium were arranged as isotropic materials and CFRP bar was set as orthorhombic anisotropic materials. The elastic modulus of the steel barrel was set to 210 GPa with a Poisson's ratio of 0.3. The mechanical parameters of CFRP bar and the bonding medium are shown in Section 3.1. The "Tie" contact mode was used for the interface between CFRP bar and the bonding medium [42,

43]. The normal direction of the contact surface between the bonding medium and the steel barrel was defined as the "hard contact" contact mode, and the tangential direction was defined as the "Coulomb friction force", and the friction coefficient was set to 0.2 [44]. Cyclic symmetry constraint was applied to the cutting surface, while fixed constraint was applied to the steel barrel, and tensile stress of 2100 MPa was applied to the CFRP cable. Each component of the anchorage was divided into hexahedral meshes, as shown in Figure 4.

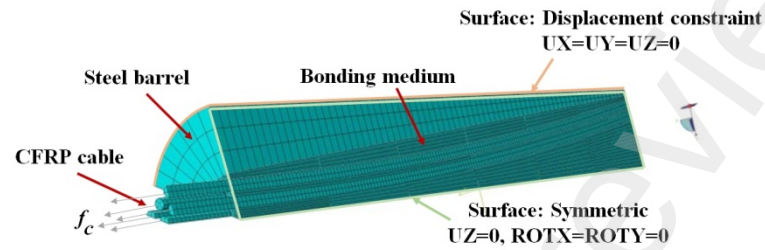


Figure 4. The mesh partition of the model ($\Phi 4-18$)

CFRP cable of $\Phi 4-18$ with four types of anchorage proposed above were simulated and analyzed, and the effect of anchorage type, bending angle of CFRP bar and bonding media on mechanical performance of anchorage was obtained. The detailed parameters used in the simulation analysis are shown in Table 2.

Table 2. Anchorage parameters of simulation analysis

Model number	Anchorage types	Angle of CFRP bars ($^{\circ}$)	Number of bonding media	E_a (GPa)	ΔE (GPa)	Length ratio of bonding media
1	EP	0	1	12	-	-
2	VP	0	6	12	4	1:1:1:1:1:1
3	EB	4	1	12	-	-
4	VB	4	6	12	4	1:1:1:1:1:1
5	VB	4	2	12	12	1:1
6	VB	4	4	12	5	1:1:1:1
7	VB	4	8	12	2	1:1:1:1:1:1:1:1
8	VB	4	6	12	4	2:2:2:1:1:1
9	VB	4	6	12	4	3:3:2:2:1:1
10	VB	4	6	12	1	1:1:1:1:1:1
11	VB	4	6	12	2	1:1:1:1:1:1
12	VB	4	6	12	3	1:1:1:1:1:1
13	VB	4	6	15	1	1:1:1:1:1:1
14	VB	4	6	15	2	1:1:1:1:1:1
15	VB	4	6	15	3	1:1:1:1:1:1
16	VB	4	6	15	4	1:1:1:1:1:1

17	VB	4	6	18	1	1:1:1:1:1:1
18	VB	4	6	18	2	1:1:1:1:1:1
19	VB	4	6	18	3	1:1:1:1:1:1
20	VB	4	6	18	4	1:1:1:1:1:1
21	VB	4	6	21	1	1:1:1:1:1:1
22	VB	4	6	21	2	1:1:1:1:1:1
23	VB	4	6	21	3	1:1:1:1:1:1
24	VB	4	6	21	4	1:1:1:1:1:1

Note: E_a : the average elastic modulus of the bonding media; ΔE : The difference between the elastic modulus of adjacent bonding media.

3 Results and discussion

3.1 Mechanical properties of materials

The tensile strength of CFRP bars under different bending angles are shown in [Table 3](#), it can be seen that the tensile strength is less than 95% of the ultimate tensile strength when the bending angle is greater than 4°. Therefore, the maximum bending angle of 4° is selected for CFRP bar in the design of anchorage. In addition, tensile elastic modulus along fiber direction, compressive elastic modulus perpendicular fiber direction and Poisson's ratio of CFRP bar with 0° bending angle are 148.45 GPa, 10.56 GPa and 0.27, respectively.

Table 3. The mechanical property parameters of CFRP bars

Bending angle (°)	0	2	4	6	8	10
Tensile strength (MPa)	2132.16	2103.48	2075.51	1997.12	1856.87	1525.02

The changes of compressive strength and elastic modulus of bonding media with mass ratio are shown in [Figure 6](#). It can be seen that the compressive strength and the elastic modulus gradually increase with mass ratio. The detailed value of compressive strength and elastic modulus are 132.54 MPa and 8.56 GPa for 1:1, 147.21 MPa and 11.24 GPa for 1:1.2, 154.74 MPa and 13.54 GPa for 1:1.4, 163.45 MPa and 14.78 GPa for 1:1.6, 171.57 MPa and 16.01 GPa for 1:1.8, 178.95 MPa and 17.28 GPa for 1:2. In addition, the difference of Poisson's ratio of bonding media for all mass ratios is negligible; the detailed value is about 0.3.

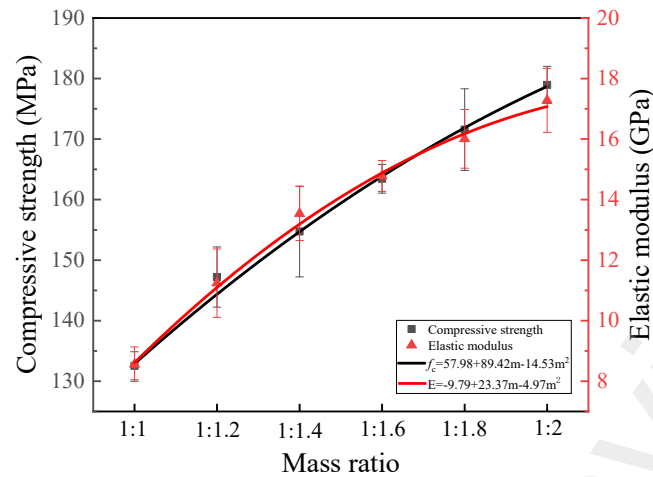


Figure 6. Changes of compressive strength and elastic modulus of bonding media with mass ratio

3.2 Tensile properties of anchorage

The load-displacement curves of two kinds of CFRP cables with four kinds of anchorages are shown in **Figure 7**, it can be seen that the load-displacement curves increase linearly in the initial loading stage, which is because the anchoring materials are in the elastic deformation stage. Subsequently, the curves show the decrease of slope or negative growth with the increase of load, which attributes to the slip between CFRP cable and anchorage. As seen from **Figure 7 (a)**, the setting of bending angle of CFRP bar can improve the bearing loading of anchorage significantly, such as, the ultimate tensile load of EB anchorage is 1.8 times more than that of EP anchorage, and the ultimate tensile load of VB anchorage is 1.7 times more than that of VP anchorage. By contrast, the improved percentage of bearing loading of anchorage by setting variable stiffness of bonding medium is lower than that of bending angle of CFRP bar. In addition, the curves of EB and VB anchorages present the sudden brittle failure, while EB and VB anchorages present the slip failure. As seen from **Figure 7 (b)**, the effect of variable stiffness of bonding medium and bending angle of CFRP bar on ultimate tensile load is insignificant, which is because the increase of anchor length improves the bearing load and avoids the slip failure of anchorage.

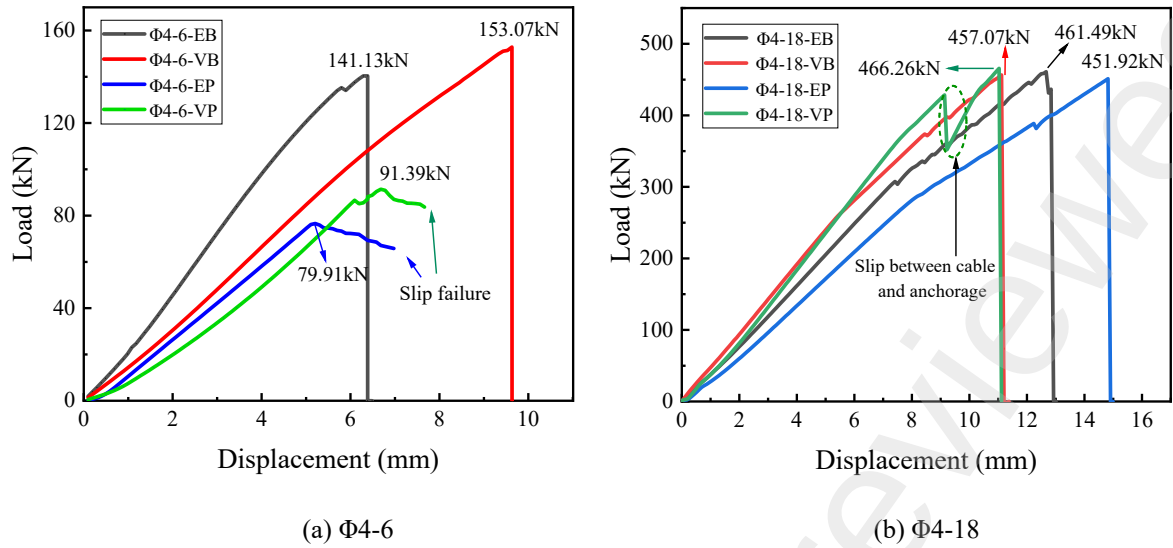


Figure 7. Load-displacement curves of CFRP cable of (a) $\Phi 4-6$ and (b) $\Phi 4-18$

Furthermore, the tensile stress-strain curves of two kinds of CFRP cables are obtained from **Figure 7** and shown in **Figure 8**. It is obvious that the stress-strain curves of CFRP cable anchored by four types of anchorages increase linearly. For VB anchorage, the tensile strength and elastic modulus of $\Phi 4-6$ cable are 2030.2 MPa and 146.9 GPa, while the tensile strength and elastic modulus of $\Phi 4-18$ cable are 2061.3 MPa and 148.3 GPa. The discreteness of tensile strength of $\Phi 4-18$ cable decreases with the increase of anchor length compared to $\Phi 4-6$ cable.

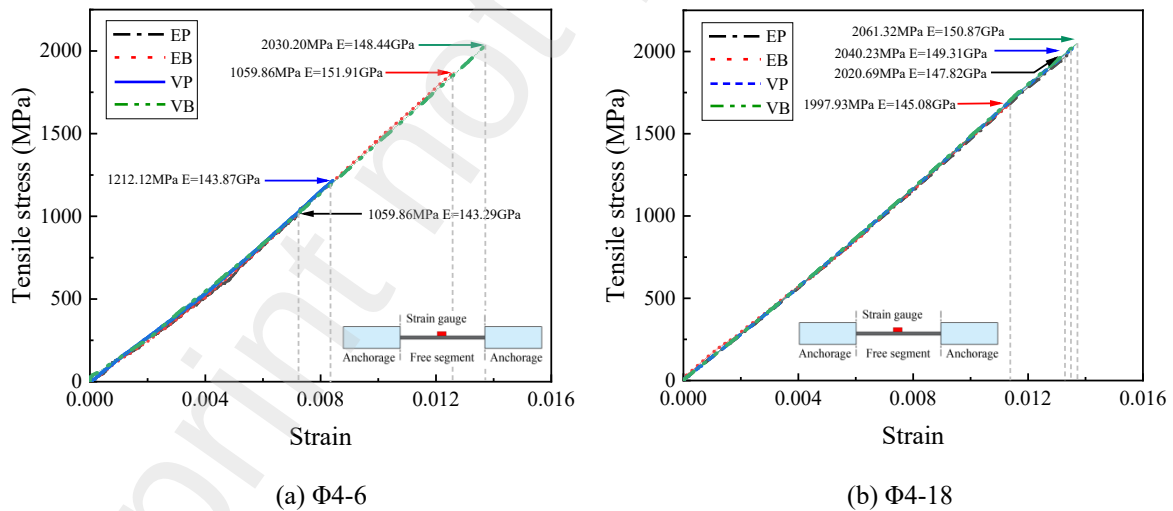


Figure 8. Tensile stress-strain curves of CFRP cables of (a) $\Phi 4-6$ and (b) $\Phi 4-18$

The three kinds of failure modes are obtained by tensile test for above four kinds of anchorages, which includes burst failure of CFRP cable, sliding failure between CFRP cable and anchorage and local shear failure of CFRP cable, as shown in **Figure 9**. The results show that EP and VP anchorages of $\Phi 4-6$ cable are prone to

slip failure, which indicated that the contact area between the parallel CFRP cable and the bonding medium is small, resulting in insufficient bonding force. In addition, EB anchorages are prone to local shear fracture, which indicates that the anchorage using a bonding medium with equal stiffness has a large stress concentration on CFRP cable at the loading end of the anchorage. All the VB anchorages present burst failure mode of CFRP cables, which indicates that the setting of variable stiffness of bonding medium and bending angle of CFRP bar not only avoids stress concentration but also improves anchoring forces.

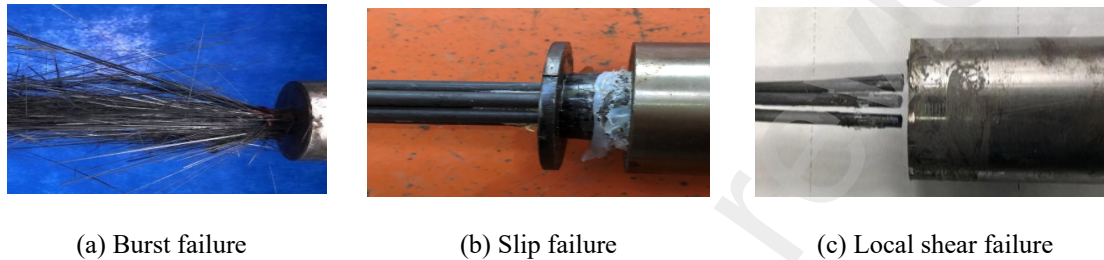


Figure 9. Tensile failure modes of CFRP cables

Based on the above results, the detailed tensile properties of anchorages are listed in **Table 5**. For $\Phi 4-6$ cables, the anchoring efficiencies of EP, VP and EB anchorages are less than 95%, while that of VB anchorage are greater than 95%, which is because the combine of variable stiffness of bonding medium and bending angle of CFRP bar can achieve higher anchoring efficiency of anchorage. In addition, anchoring efficiency of EB anchorage is higher than that of EP anchorage of both CFRP cables, which indicates that the setting of bending angle of CFRP bar can improve the bite force of bonding medium to CFRP bar. For $\Phi 4-18$ cables, the anchoring efficiency of anchorages increases significant with the increase of anchor length.

Table 5. Tensile test results of CFRP cables

CFRP cables	Failure mode	Ultimate tensile load (kN)	Anchoring efficiency (%)
$\Phi 4-6-EP$	Sliding failure	79.91	48
$\Phi 4-6-EB$	Local shear fracture	141.13	85
$\Phi 4-6-VP$	Sliding failure	91.39	55
$\Phi 4-6-VB$	Burst failure	153.07	95
$\Phi 4-18-EP$	Local shear fracture	461.49	93
$\Phi 4-18-EB$	Local shear fracture	457.07	92
$\Phi 4-18-VP$	Burst failure	451.92	95
$\Phi 4-18-VB$	Burst failure	466.26	95

3.3 Distribution of stress inside the anchorage

The compressive stress and shear stress of CFRP bar (outer layer of cable) of four types of anchorages are obtained by simulation analysis of finite element model with number of 1 to 4 (Table 2) are shown in Figure 12. It can be seen that the compressive stress and shear stress concentrations are obvious at the loading end of EP and EB anchorages, while the setting of variable stiffness of bonding medium can reduce the shear stress of VP and VB anchorages at the loading end significantly. Such as, the compressive stress at the loading end of EP, EB, VP and VB anchorages are 127.54 MPa, 181.54 MPa, 58.55 MPa and 68.95 MPa, respectively, and the shear stress at the loading end of EP, EB, VP and VB anchorages are 127.54 MPa, 181.54 MPa, 58.55 MPa and 68.95 MPa, respectively. The compressive stress and shear stress at the loading end of VB anchorage are reduced by 67.25% and 50.93% compared to EP anchorage, respectively.

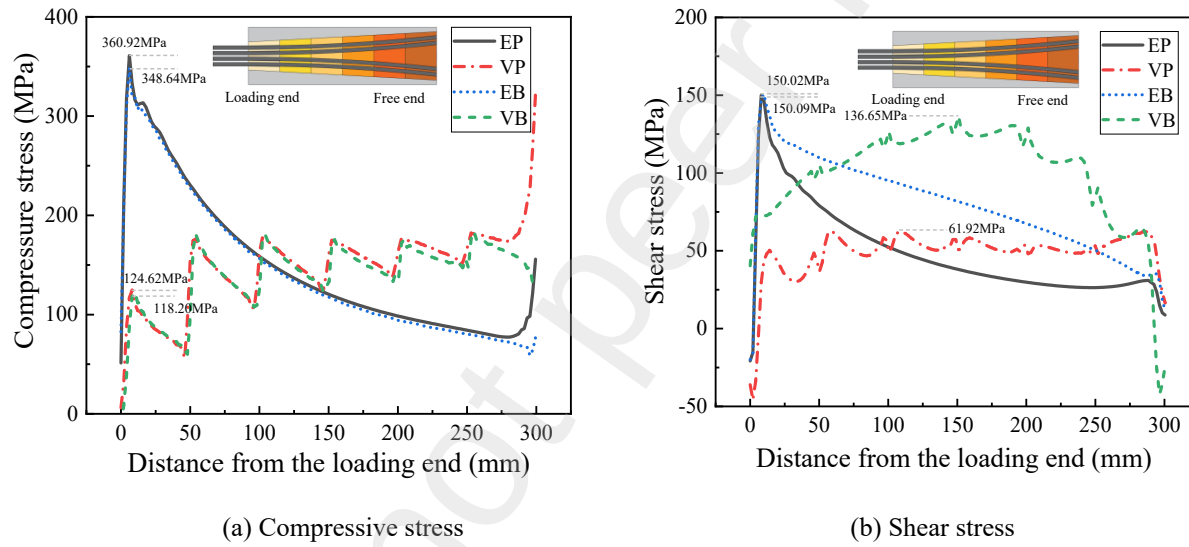


Figure 12. Distribution of stress inside the anchorage of (a) Compressive stress and (b) Shear stress for different anchorage types

Furthermore, the strains of CFRP bar (outer layer of cable) at both ends and in the middle of the anchorage zone are obtained by strain gauges, and then the distribution of shear stress inside the anchorage are calculated according to the changes of strains, which can be used to verify the validity of the simulation analysis results. The distribution of shear stress between CFRP cable and bonding medium can be described by the stress state at the dx and Δx shown in Figure 10, and the balance equation can be obtained as follows [13].

$$\tau(x) (C_f \cdot dx) = A_f \cdot (\sigma_f(x) + d\sigma_f(x)) - A_f \cdot \sigma_f(x) \quad (1)$$

Further, the shear stress can be expressed as follows.

$$\tau(x) = \frac{E_f A_f}{C_f} \cdot \frac{d\epsilon_f(x)}{dx} \quad (2)$$

The average shear stress between adjacent measuring points with a distance of Δx can be obtained as follows.

$$\bar{\tau}_i = \frac{E_f A_f}{C_f} \cdot \frac{\varepsilon_{f,i+1} - \varepsilon_{f,i}}{\Delta x} \quad (3)$$

Where: $\bar{\tau}_i$ is the average shear stress (MPa); $\varepsilon_{f,i+1}$ is the strain of CFRP cable at the $i+1$ measuring point; $\varepsilon_{f,i}$ is the strain of CFRP cable at the i measuring point; Δx is the distance between two adjacent measuring points (mm); E_f is the elastic modulus of CFRP bar (MPa); A_f is the cross-sectional area of CFRP cable (mm²); C_f is the perimeter of CFRP cable (mm); $\tau(x)$ is the shear stress at position x (MPa); $\varepsilon_f(x)$ is the strain of CFRP cable at position x ; $\sigma_f(x)$ is the stress of CFRP cable at position x (MPa).

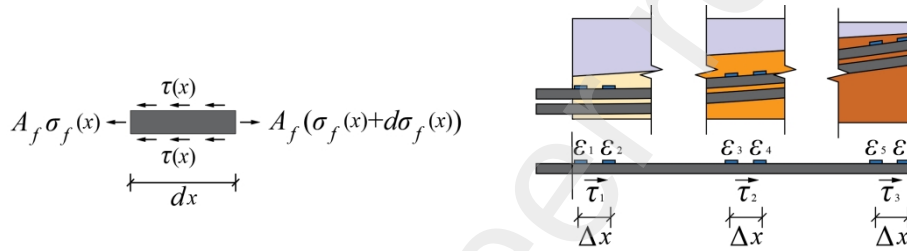


Figure 10. The distribution of shear stress calculated at the dx and Δx [13]

Through tensile test and simulation analysis, the shear strain of CFRP cables at three positions is shown in **Figure 11**. It can be seen that the setting of bending angle of CFRP bar can increase the shear stress inside the anchorage zone, especially in the middle of the anchorage zone; this was because the bending of CFRP bar increases the bite force significantly. In addition, the distribution of shear stress inside the anchorage zone obtained by the simulation analysis is consistent with that of the tensile test, and the stress difference at the loading end is 14.93%, 17.02%, 17.8% and 7.21% of the measured results, respectively. Therefore, the finite element model established in the present paper can be used to simulate the stress distribution inside the anchorage zone of CFRP cable accurately.

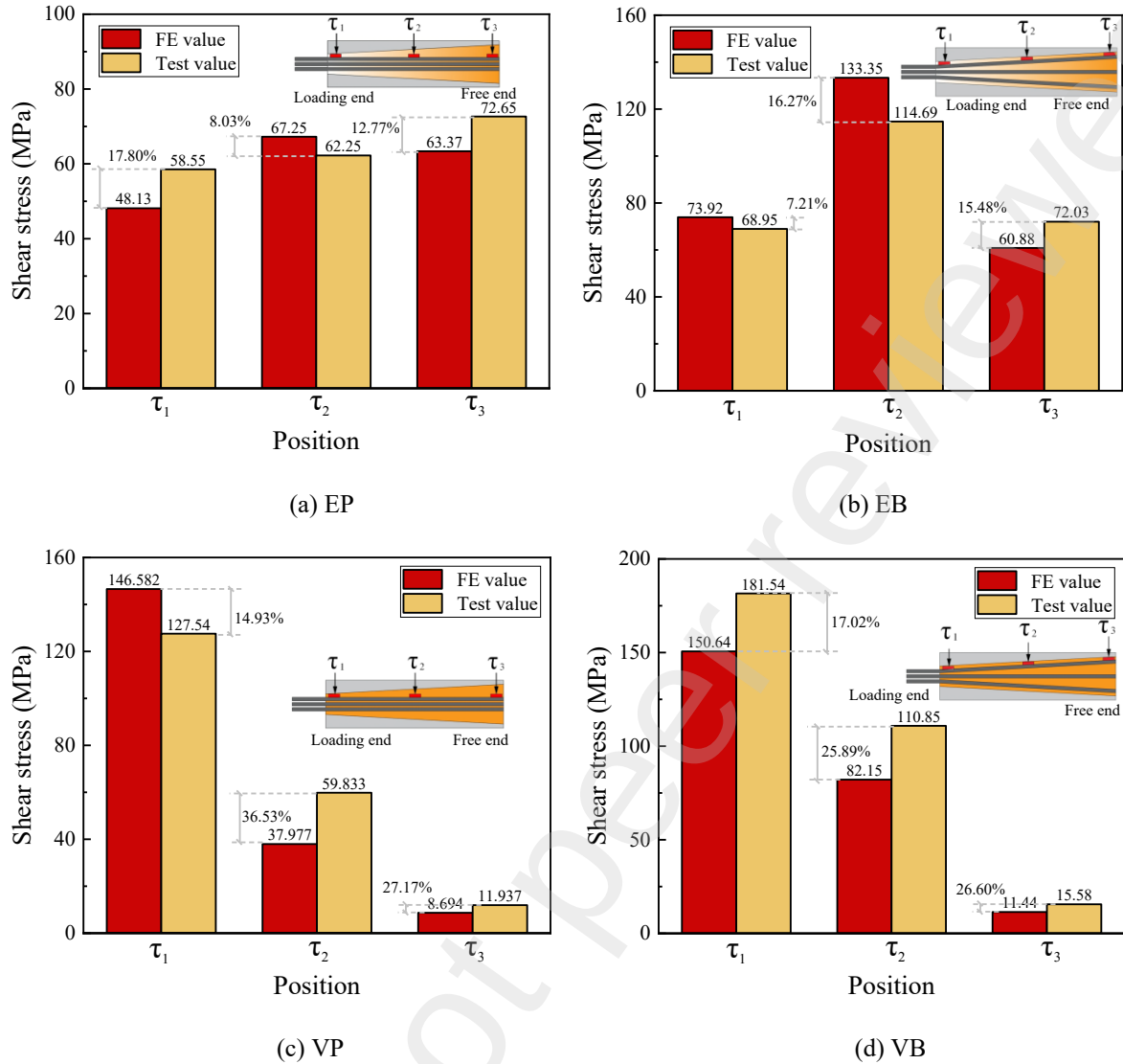


Figure 11. Shear stress of CFRP cables inside the anchorage zone of (a) EP, (b) EB, (c) VP and (d) VB anchorages

3.4 Parameter optimization of bonding medium

Based on the results and discussion of [Section 3.3](#), it can be seen that VB anchorage has the best anchorage properties. Therefore, VB anchorage is selected to conduct the parameter optimization of bonding medium. The stress distribution obtained by the finite element model with the number of 3 to 7 ([Table 2](#)) of VB anchorage under different number of bonding medium is shown in [Figure 13](#). It can be seen that the compressive stress and shear stress at the loading end of the anchorage gradually decrease with the incenses of the number of bonding medium. The compressive stress of VB anchorages with the number of bonding medium of 2, 4, 6 and 8 reduced by 22.72%, 63.52%, 64.21% and 65.22% compared to EP anchorage, respectively. In addition, the shear stress of

VB anchorages with the number of bonding medium of 2, 4, 6 and 8 reduced by 22.72%, 63.52%, 64.21% and 65.22% compared to EP anchorage, respectively. In conclusion, the number of bonding medium should be appropriately increased in the design of anchorage without increasing the difficulty for the preparation of anchorage.

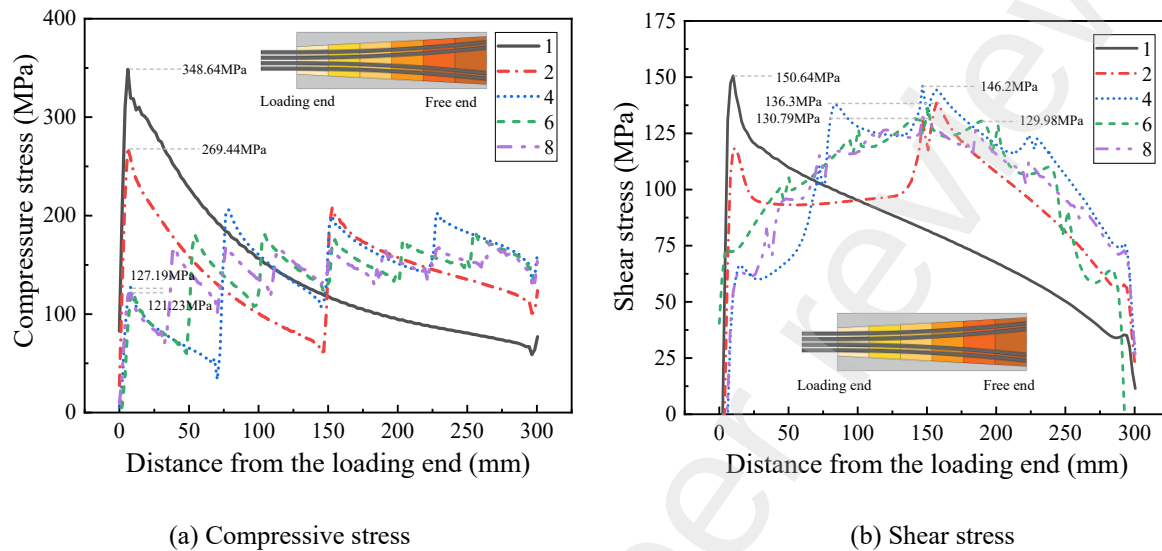


Figure 13. Distribution of stress inside the anchorage of (a) Compressive stress and (b) Shear stress for different number of bonding medium

The stress distribution obtained by the finite element model with the number of 4, 8 and 9 (Table 2) of VB anchorage under different length ratio of bonding medium is shown in Figure 14. It can be seen that the shorter the bonding media with low elastic modulus, the more uneven the distribution of compressive stress. In addition, there is no significant difference in the distribution of shear stress for the different length ratio of bonding medium. In conclusion, the length of bonding medium with different elastic modulus should be evenly distributed as far as possible in the design of anchoring.

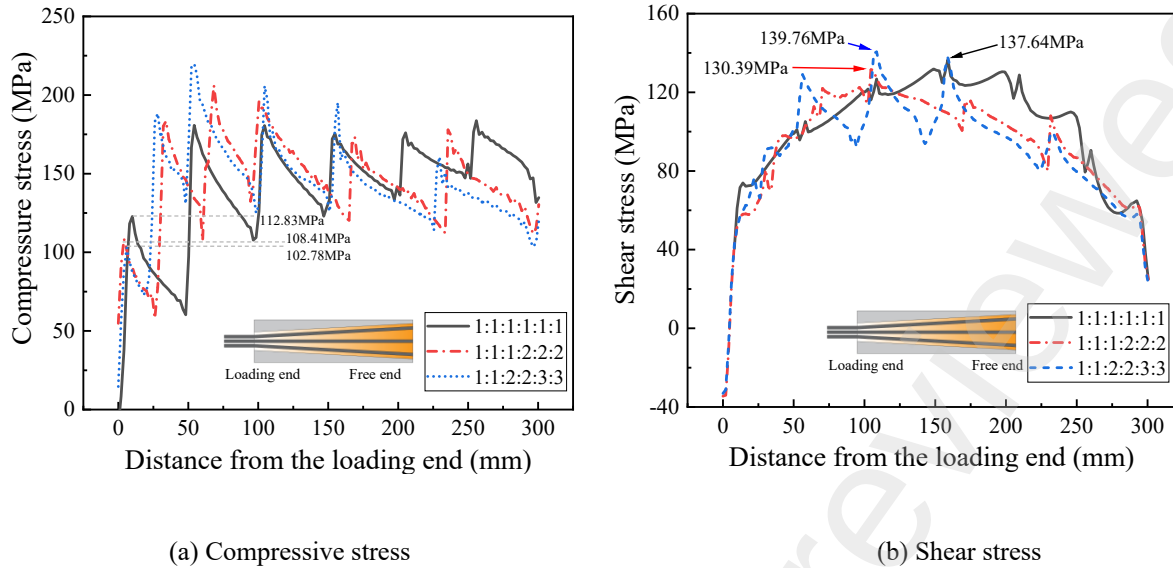


Figure 14. Distribution of stress inside the anchorage of (a) Compressive stress and (b) Shear stress for different length ratio of bonding medium

Effect of elastic modulus of bonding medium on stress peak at the loading end of VB anchoring obtained by the finite element model with the number of 4 and 10 to 24 (Table 2), as shown in Figure 15. It can be seen that with the decrease of E_a and the increase of ΔE , the peak compressive stress and shear stress at the loading end of the anchorage gradually decrease. When $E_a=4$ and $\Delta E=12$, the minimum values of shear stress and compressive stress are 118.2 MPa and 73.9 MPa, respectively. Similarly, when $E_a=1$ and $\Delta E=21$, the maximum shear stress and compressive stress are 323.5 MPa and 159.6 MPa, respectively.

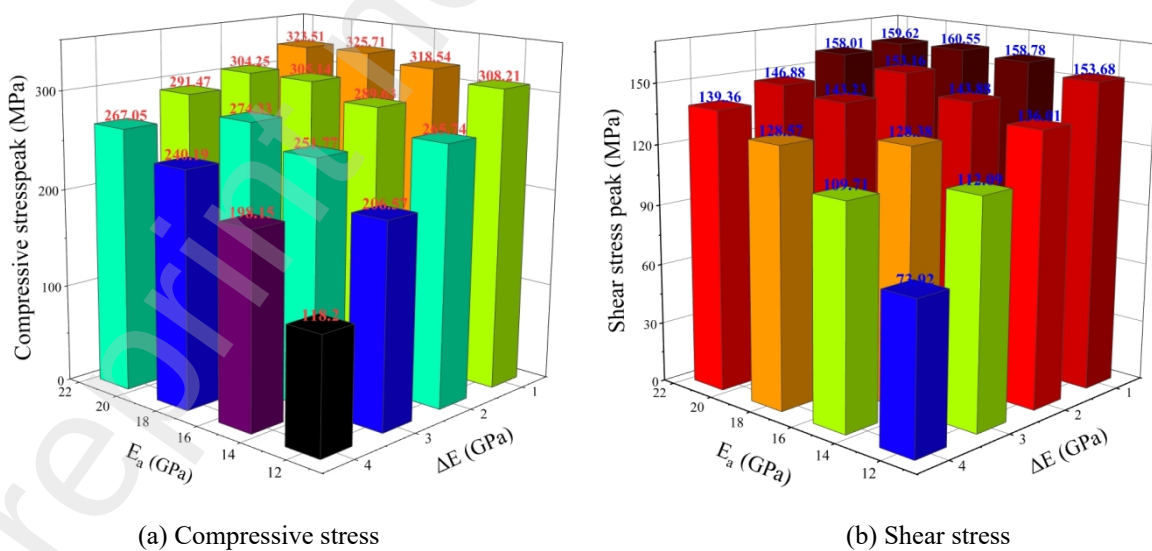


Figure 15. Effect of elastic modulus of bonding medium on stress at the loading end of (a) Compressive stress and (b) Shear stress at the loading end

4 Conclusions

In the present paper, a new type anchorage system with variable stiffness of bonding medium and bending angle of CFRP bar is developed. The design, mechanical properties, stress distribution and parameter optimization of anchorage are conducted through experiment and simulation. The following conclusions can be drawn based on the above results and discussion.

(1) The mechanical properties for anchorage design are obtained by mechanical tests, the optimal bending angle of CFRP bar is 4° .

(2) The setting of variable stiffness of bonding medium and bending angle of CFRP bar can reduce the stress peak at the loading end and increase the bite force of anchorage, which avoid the premature local shear and slip failure. The CFRP cable used anchorage with variable stiffness and bending angle (VB) anchorage presents ideal bust failure, and the anchoring efficiency is greater than 95%.

(3) The stress distribution inside the anchorage zone obtained by the simulation analysis is consistent with experimental test, and the results show that the compressive stress and shear stress at the loading end of VB anchorage are reduced by 67.3% and 50.9% compared to anchorage with equal stiffness and 0 angle (EP).

(4) Uniformity of stress distribution improved with increase of the number and the evenly of length of bonding medium, and the optimal average modulus (E_a) and difference of modulus (ΔE) of bonding medium are 12 GPa and 4 GPa.

Compliance with ethical standards

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Conflict of Interest:

The authors declare that they have no conflict of interest.

References

- [1] Y. Yang, X. Wang, Z. Wu, Long-span cable-stayed bridge with hybrid arrangement of FRP cables, *Composite Structures* 237 (2020).
- [2] Y. Liu, B. Zwingmann, M. Schlaich, Carbon Fiber Reinforced Polymer for Cable Structures-A Review, *Polymers* 7(10) (2015) 2078-2099.
- [3] Y. Wang, W. Zhang, Y. Zheng, Experimental Study on Corrosion Fatigue Performance of High-Strength Steel Wire with Initial Defect for Bridge Cable, *Applied Sciences-Basel* 10(7) (2020).
- [4] Y. Wang, Y.Q. Zheng, W.H. Zhang, Q.R. Lu, Analysis on damage evolution and corrosion fatigue performance of high-strength steel wire for bridge cable: Experiments and numerical simulation, *Theoretical and Applied Fracture Mechanics* 107 (2020).
- [5] X. Xie, X. Li, Y. Shen, Static and dynamic characteristics of a long-span cable-stayed bridge with CFRP cables, *Materials* 7(6) (2014) 4854-4877.
- [6] I. Olofin, R. Liu, The application of carbon fibre reinforced polymer (CFRP) cables in civil engineering structures, *Int. J. Civ. Eng* 2(7) (2015) 1-5.
- [7] V.M. Karbhari, Use of composite materials in civil infrastructure in Japan, WTEC report. International Technology Research Institute, World Technology (WTEC) Division (1998).
- [8] U. Meier, Proposal for a carbon fibre reinforced composite bridge across the Strait of Gibraltar at its narrowest site, *Proceedings of the Institution of Mechanical Engineers, Part B: Management and engineering manufacture* 201(2) (1987) 73-78.
- [9] U. Meier, Structural tensile elements made of advanced composite materials, *Structural engineering international* 9(4) (1999) 281-285.
- [10] M. Schlaich, Y. Liu, B. Zwingmann, Carbon Fibre Reinforced Polymer for Orthogonally Loaded Cable Net Structures, *Structural Engineering International* 25(1) (2015) 34-42.
- [11] K. Mei, Z. Lu, J. Zhang, H. Zang, Study of static load tests of CFRP stay cable anchors, *Bridge Construction* 4 (2005) 20-23.
- [12] K. Zhang, Z. Fang, A. Nanni, J. Hu, G. Chen, Experimental Study of a Large-Scale Ground Anchor System with FRP Tendon and RPC Grout Medium, *Journal of Composites for Construction* 19(4) (2015).
- [13] J. Wu, G. Xian, H. Li, A novel anchorage system for CFRP cable: Experimental and numerical investigation, *Composite Structures* 194 (2018) 555-563.
- [14] C. Li, G. Xian, Design optimization and experimental validation of a novel wedge-shaped bond anchorage system for prestressed CFRP plates, *Polymer Testing* 75 (2019) 167-174.
- [15] C. Li, R. Guo, G. Xian, H. Li, Innovative compound-type anchorage system for a large-diameter pultruded carbon/glass hybrid rod for bridge cable, *Materials and Structures* 53(4) (2020).
- [16] L. Wang, J. Zhang, J. Xu, Q. Han, Anchorage systems of CFRP cables in cable structures-A review, *Construction and Building Materials* 160 (2018) 82-99.
- [17] A. Nanni, C.E. Bakis, E.F. Oneill, T.O. Dixon, Performance of FRP tendon-anchor systems for prestressed concrete structures, *Pci Journal* 41(1) (1996) 34-&.

- [18] R. Kakihara, M. Kamiyoshi, S.-i. Kumagai, K. Noritake, A new aramid rod for the reinforcement of prestressed concrete structures, *Advanced Composites Materials in Civil Engineering Structures*, ASCE, 1991, pp. 132-142.
- [19] E.Y. Sayed-Ahmed, N.G. Shrive, A new steel anchorage system for post-tensioning applications using carbon fibre reinforced plastic tendons, *Canadian Journal of Civil Engineering* 25(1) (1998) 113-127.
- [20] A. Al-Mayah, K.A. Soudki, A. Plumtree, Experimental and analytical investigation of a stainless steel anchorage for CFRP prestressing tendons, *Pci Journal* 46(2) (2001) 88-100.
- [21] M.M.R. Taha, N.G. Shrive, New concrete anchors for carbon fiber-reinforced polymer post-tensioning tendons—Part 1: State-of-the-art review/design, *Structural Journal* 100(1) (2003) 86-95.
- [22] L.A. Lutz, P. Gergely, Mechanics of bond and slip of deformed bars in concrete, *Journal Proceedings*, 1967, pp. 711-721.
- [23] A. Katz, Bond mechanism of FRP rebars to concrete, *Materials and structures* 32 (1999) 761-768.
- [24] B.R. Zhang, B. Benmokrane, Design and evaluation of a new bond-type anchorage system for fiber reinforced polymer tendons, *Canadian Journal of Civil Engineering* 31(1) (2004) 14-26.
- [25] R. Wolff, H.-J. Miessler, New materials for prestressing and monitoring heavy structures, *Concrete International* 11(9) (1989) 86-89.
- [26] S. Mochida, T. Tanaka, K. Yagi, The development and application of a ground anchor using new materials, *Proc., 1st Int. Conf. on Advanced Compos. Mat. in Bridges and Struct*, Canadian Society for Civil Engineering, Montreal, 1992, pp. 393-402.
- [27] U. Meier, H. Meier, P. Kim, Anchorage device for high-performance fiber composite cables, *Google Patents*, 1998.
- [28] T. Jiang, Z. Fang, Experimental investigation on the performance of wedge-bond anchors for CFRP tendons, *Chi. Civ. Eng. J* 43 (2010) 79-87.
- [29] M. Chen, G. Chen, Z. Fang, K. Zhang, J. Hu, R. Liu, Large-scale ground anchorage system with high performance materials, *Proc., 28th Annual Int. Bridge Conf., Engineers' Society of Western Pennsylvania*, Pittsburgh, 2011, pp. 694-699.
- [30] D.-s. Cai, J. Yin, R.-g. Liu, Experimental and analytical investigation into the stress performance of composite anchors for CFRP tendons, *Composites Part B-Engineering* 79 (2015) 530-534.
- [31] E.A.W. Menezes, L.V. Silva, R.J. Marczak, S.C. Amico, Numerical model updating applied to the simulation of carbon fiber-reinforced polymer cables under bending and tensile stress, *Journal of Strain Analysis for Engineering Design* 52(6) (2017) 356-364.
- [32] G.-h. Xie, J. Yin, R.-g. Liu, B. Chen, D.-s. Cai, Experimental and numerical investigation on the static and dynamic behaviors of cable-stayed bridges with CFRP cables, *Composites Part B-Engineering* 111 (2017) 235-242.
- [33] R.A. Mitchell, R.M. Woolley, N. Halsey, High-strength end fittings for FRP rod and rope, *Journal of the Engineering Mechanics Division* 100(4) (1974) 687-706.
- [34] A. Nanni, C.E. Bakis, E.F. Oneil, T.O. Dixon, Short-term sustained loading of FRP tendon-anchor systems, *Construction and Building Materials* 10(4) (1996) 255-266.
- [35] T.-K. Kim, W.-T. Jung, Improvement of anchorage performance of carbon fiber-reinforced polymer cables, *Polymers* 14(6) (2022) 1239.

- [36] J.-J. Zheng, J.-G. Dai, Prediction of the nonlinear pull-out response of FRP ground anchors using an analytical transfer matrix method, *Engineering Structures* 81 (2014) 377-385.
- [37] G.-h. Xie, Y.-s. Tang, C.M. Wang, S.-q. Li, R.-g. Liu, Experimental study on fatigue performance of adhesively bonded anchorage system for CFRP tendons, *Composites Part B-Engineering* 150 (2018) 47-59.
- [38] Y.M. Saeed, S.M. Al-Obaidi, E.G. Al-hasany, F.N. Rad, Evaluation of a new bond-type anchorage system with expansive grout for a single FRP rod, *Construction and Building Materials* 261 (2020).
- [39] B. Feng, J. Zhong, H. Li, An Analytical Research about Parameter Influence on Large-Scale Anchorage Region with Multiple CFRP Cables, *Ksce Journal of Civil Engineering* 25(2) (2021) 540-551.
- [40] J. Zhou, X. Wang, Z. Peng, Z. Wu, Z. Zhu, Evaluation of a large-tonnage FRP cable anchor system: Anchorage design and full-scale experiment, *Engineering Structures* 251 (2022).
- [41] Z. Wang, R. Guo, P.X. Zhang, J.J. Shi, C.G. Li, B. Hong, G.J. Xian, Transverse low-velocity impact behaviors of pultruded carbon-fiber-reinforced polymer rods with tensile preloads: Experiment and simulation, *Compos. Pt. B-Eng.* 283 (2024) 14.
- [42] J. Noisternig, D. Jungwirth, Design and analysis of anchorage systems for a carbon fiber composite cable, *Proc., 2nd Int. Conf. on Advanced Compos. Mat. in Bridges and Struct*, Canadian Society for Civil Engineering, Montreal, 1996, pp. 935-942.
- [43] D.-s. Cai, Z.-h. Xu, J. Yin, R.-g. Liu, G. Liang, A numerical investigation on the performance of composite anchors for CFRP tendons, *Construction and Building Materials* 112 (2016) 848-855.
- [44] A. Al-Mayah, K. Soudki, A. Plumtree, Novel anchor system for CFRP rod: Finite-element and mathematical models, *Journal of Composites for Construction* 11(5) (2007) 469-476.